

PRAKTIKUM SISTEM OPERASI

MODUL IV

MEMBACA SOURCE CODE XINU



Disusun Oleh :

Alya Rabani / 2311104076

SE-07-02

Dosen Pengampu :

Yudha Islami Sulistya, S.Kom., M.Cs.

PROGRAM STUDI S1 SOFTWARE ENGINEERING

FAKULTAS INFORMATIKA

TELKOM UNIVERSITY PURWOKERTO

2025

JURNAL

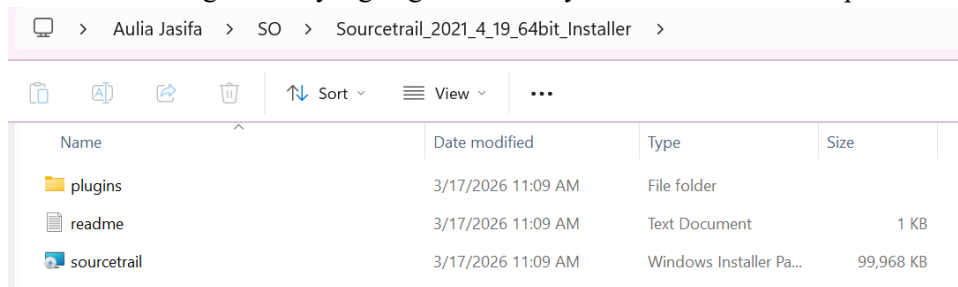
1. [10 Poin] Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

Hint: baca kembali modul-modul sebelumnya

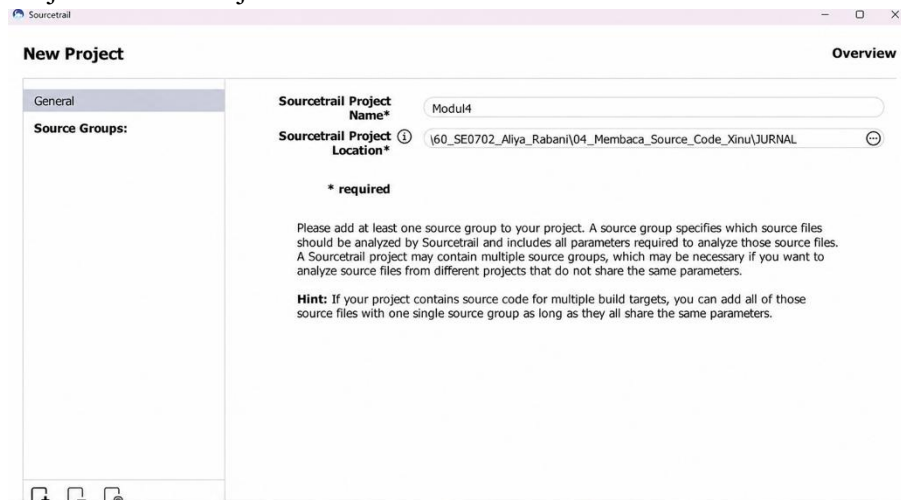
Jawaban: **Nama image:** biasanya bernama xinu.elf atau xinu.bin (tergantung konfigurasi build). **Lokasi file:** berada di folder: xinu/compile/. **Ukuran file:** bervariasi ($\pm$ ratusan KB, tergantung build & konfigurasi)

2. Membaca source code Xinu

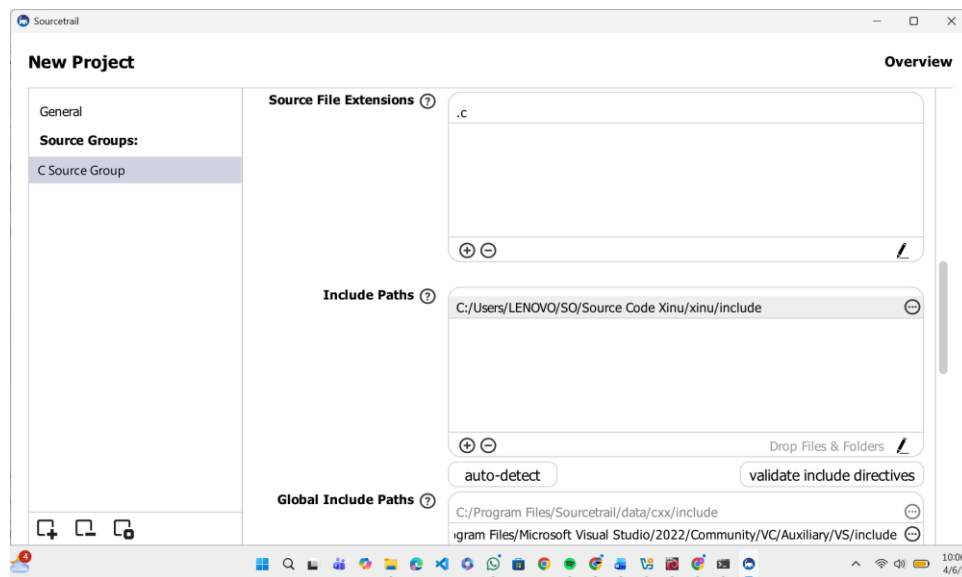
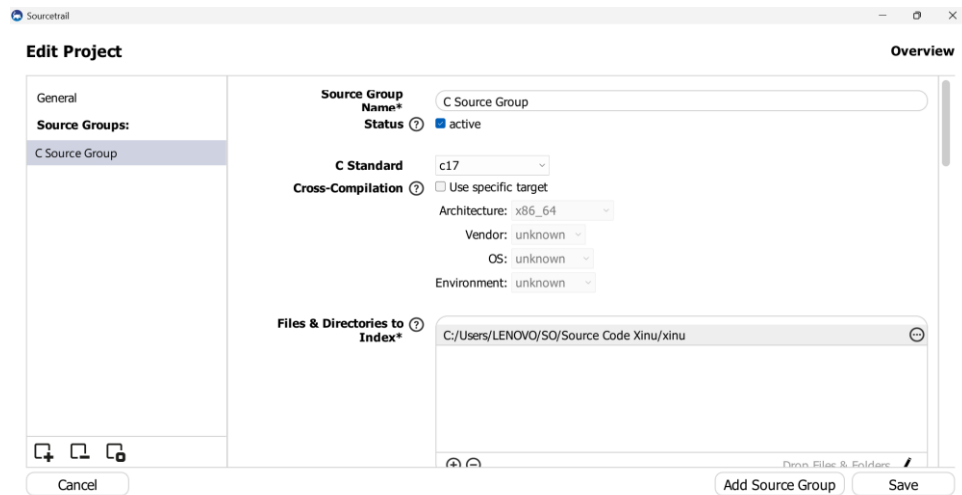
- a. Cek aplikasi bernama Sourcetrail di PC. Jika belum ada, download SourceTrail pada (DOWNLOAD SOURCETRAIL). SourceTrail adalah software untuk mengeksplorasi source code. Programmer yang bagus lebih banyak membaca kode daripada menulis kode



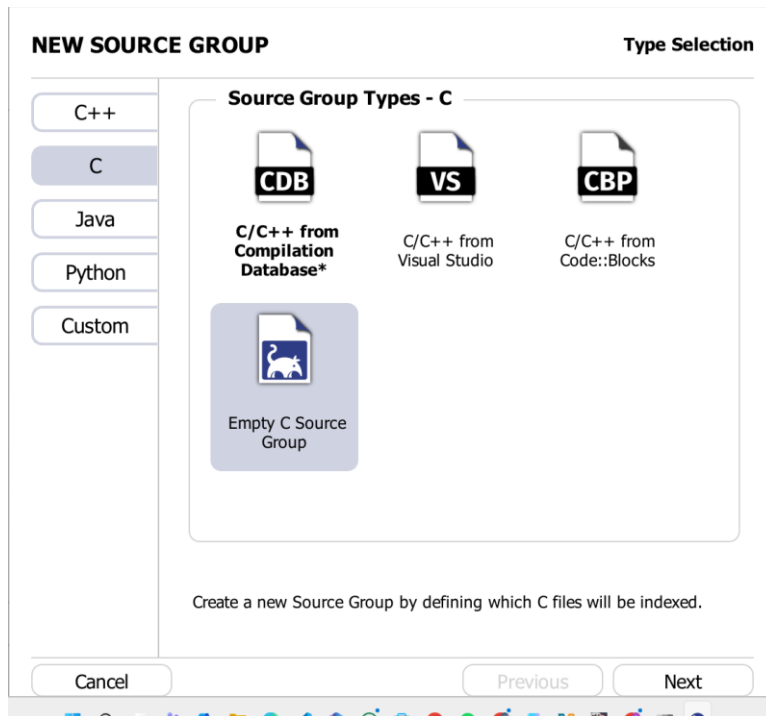
- b. Jalankan SourceTrail
- c. Project → New Project



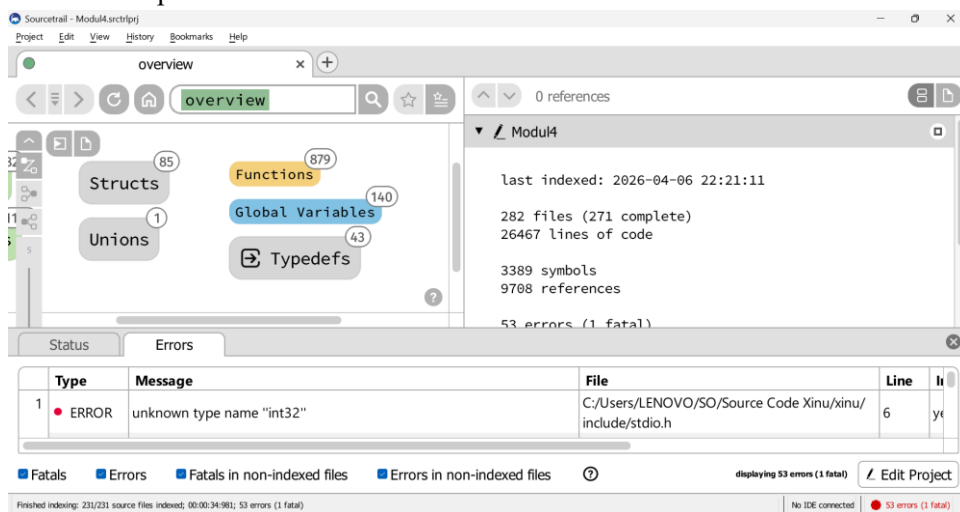
- d. Isi nama project xinu dan pilih lokasi project di manapun



- e. Add Source Groups, pilih C, lalu pilih Empty C Source Group



- f. File & Directories to Index: masukkan semua folder Xinu (yang sebelumnya telah di download)
- g. Include Paths: ../xinu/include
- h. Create
- i. Silahkan eksplorasi source code Xinu



SourceTrail - Modul4.scrtrprj

Project Edit View History Bookmarks Help

rdspcress.c

1 references

rdspcress.c

```

1 /* rdspcress.c - rdspcress */
2
3 #include <xinu.h>
4
5 /*-----
6  * rdspcress - High-priority background process tha
7  * extracts an item from the request queue, sen
8  * request to the remote disk server, and handl
9  * response, including caching responses blocks

```

Status Errors

Type	Message	File	Line	Info
1 ERROR	unknown type name "int32"	C:/Users/LENOVO/SO/Source Code Xinu/xinu/include/stdio.h	6	ye

Fatals Errors Fatals in non-indexed files Errors in non-indexed files displaying 53 errors (1 fatal) Edit Project

Activate "C:/Users/LENOVO/SO/Source Code Xinu/xinu/device/rdspcress.c": 1 result with 0 references in 1 file No IDE connected 53 errors (1 fatal)

SourceTrail - Modul4.scrtrprj

Project Edit View History Bookmarks Help

create.c

1 references

create.c

```

1 /* create.c - create, newpid */
2
3 #include <xinu.h>
4
5 local int newpid();
6
7 /*-----
8  * create - Create a process to start running a fun
9  *

```

Status Errors

Type	Message	File	Line	Info
1 ERROR	unknown type name "int32"	C:/Users/LENOVO/SO/Source Code Xinu/xinu/include/stdio.h	6	ye

Fatals Errors Fatals in non-indexed files Errors in non-indexed files displaying 53 errors (1 fatal) Edit Project

Activate "C:/Users/LENOVO/SO/Source Code Xinu/xinu/system/create.c": 1 result with 0 references in 1 file No IDE connected 53 errors (1 fatal)

SourceTrail - Modul4.scrtrprj

Project Edit View History Bookmarks Help

kill.c

1 references

kill.c

```

1 /* kill.c - kill */
2
3 #include <xinu.h>
4
5 /*-----
6  * kill - Kill a process and remove it from the syst
7  *
8  */
9 syscall kill(

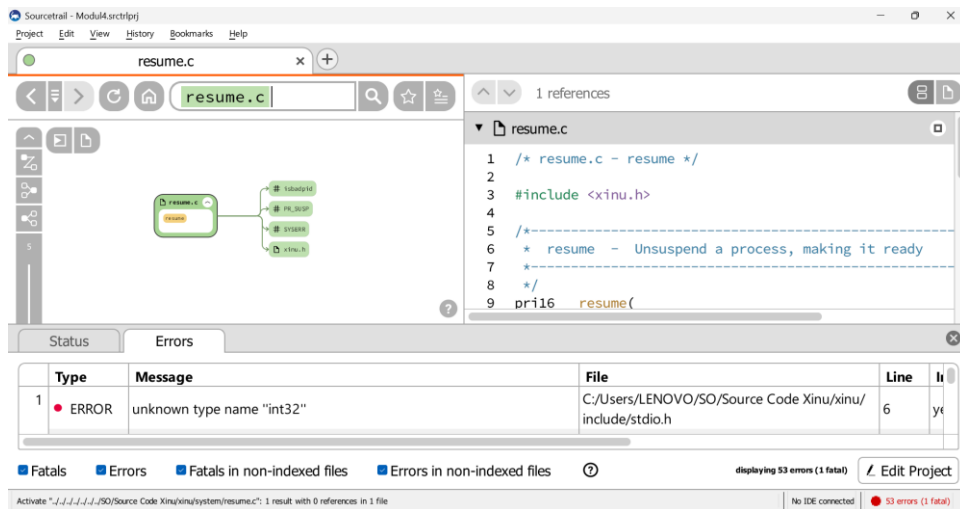
```

Status Errors

Type	Message	File	Line	Info
1 ERROR	unknown type name "int32"	C:/Users/LENOVO/SO/Source Code Xinu/xinu/include/stdio.h	6	ye

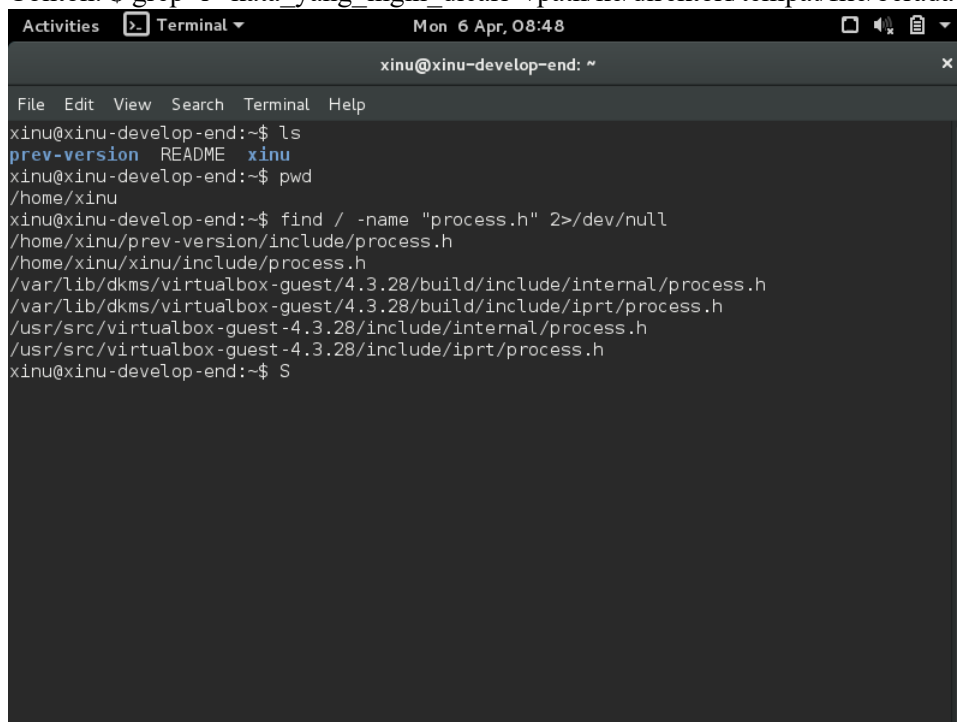
Fatals Errors Fatals in non-indexed files Errors in non-indexed files displaying 53 errors (1 fatal) Edit Project

Activate "C:/Users/LENOVO/SO/Source Code Xinu/xinu/system/kill.c": 1 result with 0 references in 1 file No IDE connected 53 errors (1 fatal)



3. [10 Poin] Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut? Hint: file berektensi .h Pada Linux bisa digunakan perintah "grep"

Contoh: `$ grep -r "kata_yang_ingin_dicari" /path/ke/direktori/tempat/file/berada`



```
Activities Terminal Mon 6 Apr, 08:53
xinu@xinu-develop-end: ~

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ ls
prev-version  README  xinu
xinu@xinu-develop-end:~$ pwd
/home/xinu
xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/include
/home/xinu/xinu/include/shell.h:/* Shell banner (assumes VT100) */
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/shell/
/home/xinu/xinu/shell/shell.c: /* Print shell banner and startup message */
xinu@xinu-develop-end:~$ grep -r "struct procent" /home/xinu/xinu/include/
/home/xinu/xinu/include/process.h:struct procent { /* Entry in the process
table */
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      8
#endif

/* Process state constants */
```

```
Activities Terminal Mon 6 Apr, 08:53
xinu@xinu-develop-end: ~

File Edit View Search Terminal Help
/* Process state constants */

#define PR_FREE      0 /* Process table entry is unused */
#define PR_CURR      1 /* Process is currently running */
#define PR_READY     2 /* Process is on ready queue */
#define PR_RECV      3 /* Process waiting for message */
#define PR_SLEEP     4 /* Process is sleeping */
#define PR_SUSP      5 /* Process is suspended */
#define PR_WAIT      6 /* Process is on semaphore queue */
#define PR_RECTIM    7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN        16 /* Length of process "name" */
#define NULLPROC      0 /* ID of the null process */

/* Process initialization constants */

#define INITSTK        65536 /* Initial process stack size */
#define INITPRI0       20 /* Initial process priority */
#define INITRET        userret /* Address to which process returns */

/* Inline code to check process ID (assumes interrupts are disabled) */

#define isbadpid(x)    ( ((pid32)(x) < 0) || \
                        ((pid32)(x) >= NPROC) || \
                        (proctab[(x)].prstate == PR_FREE))
```

4. [80 poin] Mengubah welcome banner pada Xinu
- Carilah file yang menyimpan banner Xinu!
Hint: file berekstensi .h pada direktori xinu/include
 - Carilah file yang menampilkan banner Xinu!
Hint: file berekstensi .c pada direktori xinu/shell
 - Ubahlah welcome banner Xinu sehingga menjadi seperti ini:

```

Activities [?] Terminal Tue 7 Apr, 07:46
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: include/shell.h Modified

/* Shell banner (assumes VT100) */

#define SHELL_BAN0 "\033[31;1m"
#define SHELL_BAN1 "
#define SHELL_BAN2 "-----ALYA RABANI-----"
#define SHELL_BAN3 "
#define SHELL_BAN4 "
#define SHELL_BAN5 "
#define SHELL_BAN6 "
#define SHELL_BAN7 "
#define SHELL_BAN8 "-----2311104076-----"
#define SHELL_BAN9 "\033[0m\n"

/* Messages shell displays for user */

#define SHELL_PROMPT "xsh $ " /* Command prompt */
#define SHELL_STRMSG "Welcome to Xinu!\n"/* Welcome message */
#define SHELL_EXITMSG "Shell closed\n"/* Shell exit message */
#define SHELL_SYNERMSG "Syntax error\n"/* Syntax error message */
#define SHELL_CREATMSG "Cannot create process\n"/* command error */
#define SHELL_INERMSG "Cannot open file %s for input\n"/* Input err */
#define SHELL_OUTERMSG "Cannot open file %s for output\n"/* Output err */

CG Get Help  CO WriteOut  CR Read File  CV Prev Page  CX Cut Text  CC Cur Pos
CX Exit  CJ Justify  CW Where Is  CN Next Page  CU UnCut Text  CT To Spell

```

- ```
Activities Terminal Mon 13 Apr, 20:48
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
4447448 bytes of free memory. *Free list:
[0x0015E320 to 0x009FFF7]
[0x00100000 to 0x005FBFFF]
115657 bytes of Xinu code.
[0x00100000 to 0x0011C3C8]
148488 bytes of data.
[0x0011FEC0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0xa08030f)

-----ALYA RABANI-----

XINU

-----2311104076-----

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```